

SNAPDRAGON® AI LAB BUILD & PRESENT CHALLENGE

SmartQC

On-device visual quality inspection, built and optimized for Snapdragon.

Samay Thakur

Electronics & Instrumentation Engineering, M.S. Ramaiah Institute of Technology, Bengaluru

[GitHub](#) | [Linkedin](#)

THE PROBLEM

Quality inspection is slow, inconsistent, or expensive

1

Manual inspection

Depends on human attention span — fatigue and inconsistency creep in over a shift.

2

Cloud-based AI tools

Solve accuracy, but add latency, an internet dependency, and data-privacy exposure.

3

Industrial inspection hardware

Accurate and fast, but priced far out of reach for small food producers and workshops.

There is a need for a fast, private, self-contained inspection tool that runs entirely on the device performing the inspection.

THE SOLUTION

SmartQC: factory-grade inspection, without the factory-grade price tag

WHO IT'S FOR

Small bakeries, food producers, and workshops that need reliable visual QC but can't justify a dedicated inspection line.

WHAT IT DOES

Classifies product quality from a webcam or uploaded image, entirely on-device — no cloud calls, no internet dependency at inference time.

WHY IT MATTERS

Runs on a Snapdragon laptop the business may already own — zero extra hardware cost.

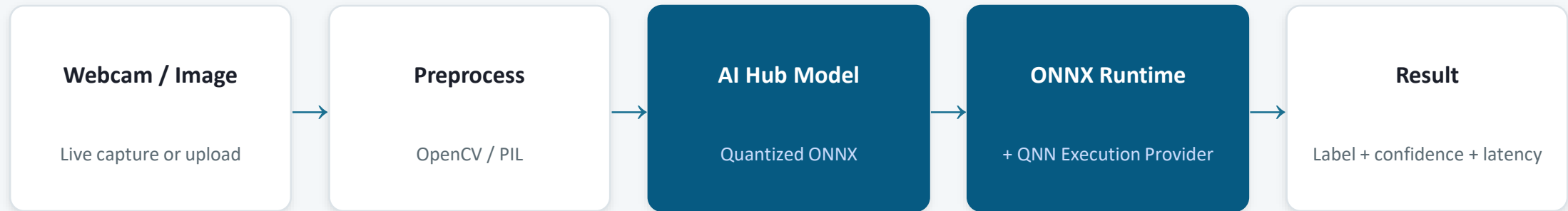
VALUE PROPOSITION

“Factory-grade visual inspection, without the factory-grade price tag.”

Built by extending an earlier ESP32-CAM + MobileNetV2 bread-quality system into a fully software-based Windows app for Snapdragon-powered laptops.

HOW IT WORKS

Architecture



The optimized model and ONNX Runtime + QNN Execution Provider path are what make this pipeline Snapdragon-NPU-ready.

App features: live webcam / upload inspection • batch folder inspection + CSV export • confidence threshold flagging • local SQLite history log • rolling defect-rate trend chart • fully offline at inference time

WHY ON-DEVICE, WHY SNAPDRAGON

Built for where inspection actually happens

Cloud-based inspection

- ✗ Needs constant internet
- ✗ Per-image / API cost
- ✗ Data leaves the device
- ✗ Latency varies with connection

SmartQC on Snapdragon

- ✓ Fully offline at inference
- ✓ Zero marginal cost per scan
- ✓ All data stays on-device
- ✓ NPU-optimized, low-latency

Snapdragon's on-device NPU + long battery life means SmartQC can run all day on a shop floor — unplugged, private, and instant.

MODEL TRAINING

MobileNetV2, fine-tuned to 99.2% validation accuracy

99.2%

Best validation accuracy

649

Total labeled images (Fresh / Mold bread)

12

Training epochs

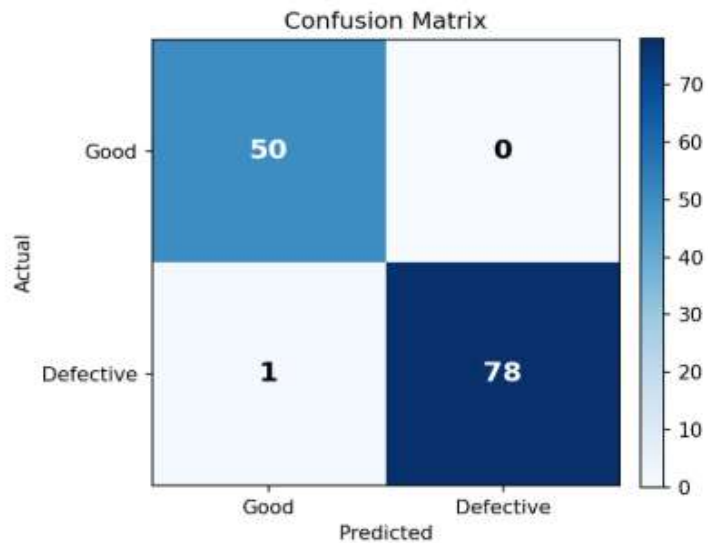
```
● (venv) PS C:\Users\samay\OneDrive\Desktop\SmartQC\app> cd ..
● (venv) PS C:\Users\samay\OneDrive\Desktop\SmartQC> python train.py
Found classes: ['Fresh Bread(Good Bread) Images', 'Mold Bread(Bad Bread) Images']
Total images: 649
Train: 520 Val: 129
Epoch 1/12 train_loss=0.2621 train_acc=0.902 val_acc=0.984
Epoch 2/12 train_loss=0.1234 train_acc=0.956 val_acc=0.992
Epoch 3/12 train_loss=0.0647 train_acc=0.977 val_acc=0.992
Epoch 4/12 train_loss=0.0595 train_acc=0.981 val_acc=0.984
Epoch 5/12 train_loss=0.0374 train_acc=0.992 val_acc=0.984
Epoch 6/12 train_loss=0.0208 train_acc=0.994 val_acc=0.984
Epoch 7/12 train_loss=0.0169 train_acc=0.994 val_acc=0.977
Epoch 8/12 train_loss=0.0135 train_acc=0.996 val_acc=0.984
Epoch 9/12 train_loss=0.0538 train_acc=0.992 val_acc=0.984
Epoch 10/12 train_loss=0.0698 train_acc=0.977 val_acc=0.984
Epoch 11/12 train_loss=0.0220 train_acc=0.990 val_acc=0.969
Epoch 12/12 train_loss=0.0202 train_acc=0.996 val_acc=0.984

Best validation accuracy: 0.992
```

Fine-tuned on a Fresh Bread (Good) / Mold Bread (Defective) dataset using transfer learning from ImageNet-pretrained MobileNetV2 — the same architecture family used in the original bread-quality-indicator project this builds on.

MODEL EVALUATION

Beyond accuracy: confusion matrix, precision/recall, ROC-AUC

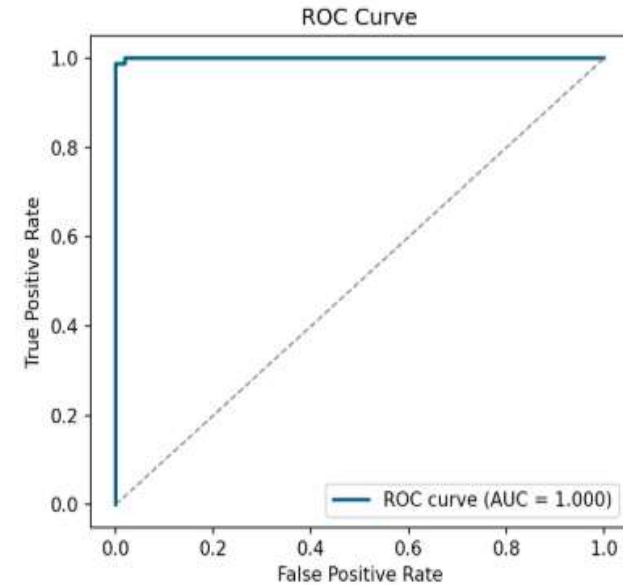


129

Validation images

1

Total misclassifications



0.9997

ROC-AUC

Compiled and optimized for the Snapdragon X Elite

COMPILE

QUANTIZE

LINK

PROFILE

INFERENCE

Creator: samaythakur999@gmail.com (me)

Rows per page: 10

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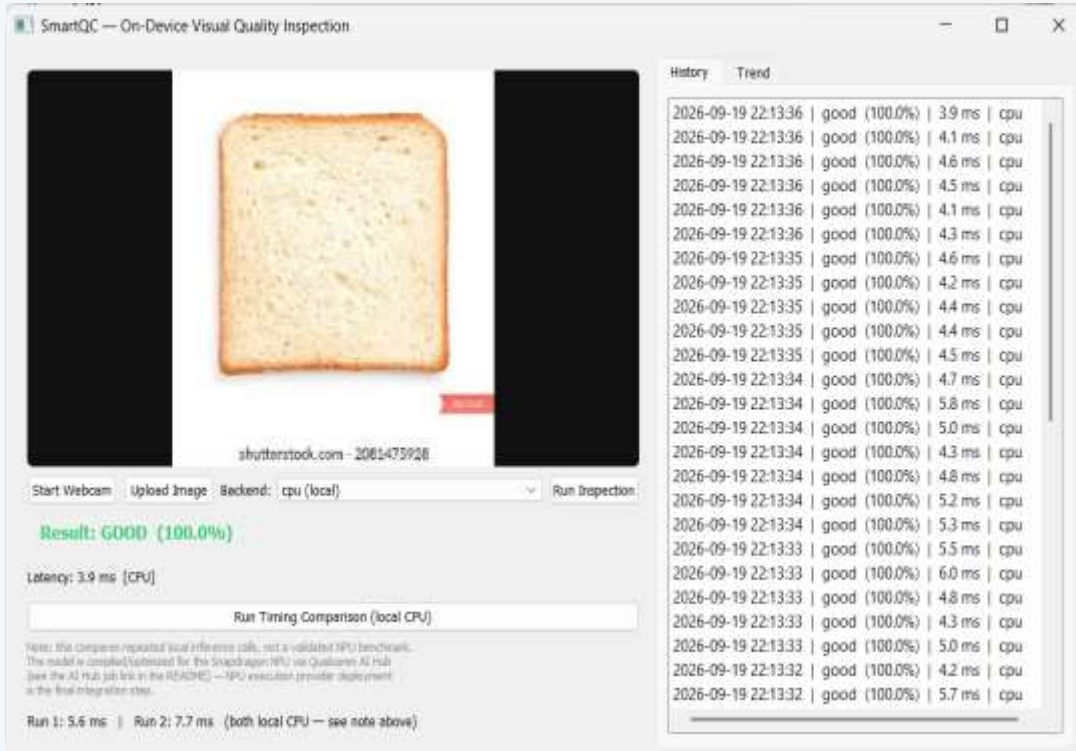
>>

Job Name	Job ID	Status	Target Device	Creator	Submission Time
smartqc_model.onnx	jpe782zl5	Results Ready	Snapdragon X Elite CRD	samaythakur999@gmail.com	9/20/2026, 2:51:55 AM

Compile job jpe782z15 • Target: Snapdragon X Elite CRD • Status: Results Ready

LIVE DEMO

Correctly classifies both good and defective samples



SmartQC — On-Device Visual Quality Inspection

History Trend

2026-09-19 22:13:36	good (100.0%)	3.9 ms	cpu
2026-09-19 22:13:36	good (100.0%)	4.1 ms	cpu
2026-09-19 22:13:36	good (100.0%)	4.6 ms	cpu
2026-09-19 22:13:36	good (100.0%)	4.5 ms	cpu
2026-09-19 22:13:36	good (100.0%)	4.1 ms	cpu
2026-09-19 22:13:36	good (100.0%)	4.3 ms	cpu
2026-09-19 22:13:35	good (100.0%)	4.6 ms	cpu
2026-09-19 22:13:35	good (100.0%)	4.2 ms	cpu
2026-09-19 22:13:35	good (100.0%)	4.4 ms	cpu
2026-09-19 22:13:35	good (100.0%)	4.4 ms	cpu
2026-09-19 22:13:35	good (100.0%)	4.5 ms	cpu
2026-09-19 22:13:34	good (100.0%)	4.7 ms	cpu
2026-09-19 22:13:34	good (100.0%)	5.8 ms	cpu
2026-09-19 22:13:34	good (100.0%)	5.0 ms	cpu
2026-09-19 22:13:34	good (100.0%)	4.3 ms	cpu
2026-09-19 22:13:34	good (100.0%)	4.8 ms	cpu
2026-09-19 22:13:34	good (100.0%)	5.2 ms	cpu
2026-09-19 22:13:34	good (100.0%)	5.3 ms	cpu
2026-09-19 22:13:33	good (100.0%)	5.5 ms	cpu
2026-09-19 22:13:33	good (100.0%)	6.0 ms	cpu
2026-09-19 22:13:33	good (100.0%)	4.8 ms	cpu
2026-09-19 22:13:33	good (100.0%)	4.3 ms	cpu
2026-09-19 22:13:33	good (100.0%)	5.0 ms	cpu
2026-09-19 22:13:32	good (100.0%)	4.2 ms	cpu
2026-09-19 22:13:32	good (100.0%)	5.7 ms	cpu

Result: **GOOD (100.0%)**

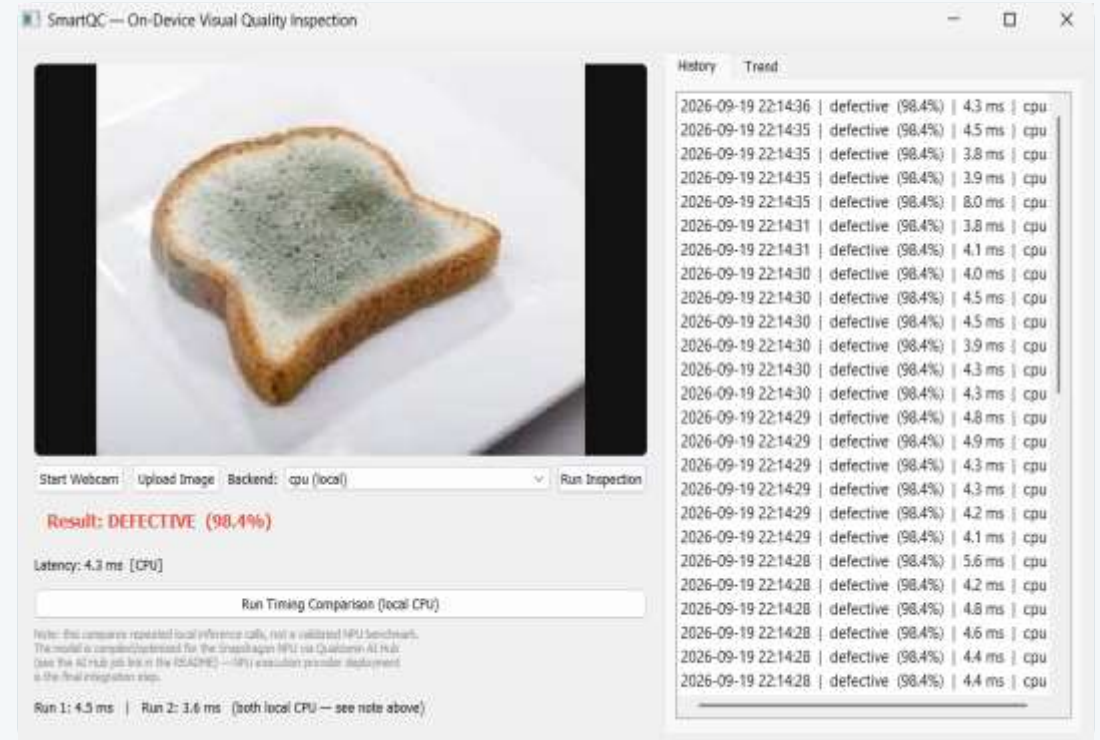
Latency: 3.9 ms [CPU]

Run Timing Comparison (local CPU)

Run 1: 5.6 ms | Run 2: 7.7 ms (both local CPU — see note above)

Note: this compares repeated local inference calls, not a validated NPU benchmark. The model is compiled/optimized for the Snapdragon NPU via Qualcomm AI Hub (see the AI Hub job link in the README) — NPU execution provider deployment is the final integration step.

GOOD — 100.0% confidence



SmartQC — On-Device Visual Quality Inspection

History Trend

2026-09-19 22:14:36	defective (98.4%)	4.3 ms	cpu
2026-09-19 22:14:35	defective (98.4%)	4.5 ms	cpu
2026-09-19 22:14:35	defective (98.4%)	3.8 ms	cpu
2026-09-19 22:14:35	defective (98.4%)	3.9 ms	cpu
2026-09-19 22:14:35	defective (98.4%)	8.0 ms	cpu
2026-09-19 22:14:31	defective (98.4%)	3.8 ms	cpu
2026-09-19 22:14:31	defective (98.4%)	4.1 ms	cpu
2026-09-19 22:14:30	defective (98.4%)	4.0 ms	cpu
2026-09-19 22:14:30	defective (98.4%)	4.5 ms	cpu
2026-09-19 22:14:30	defective (98.4%)	4.5 ms	cpu
2026-09-19 22:14:30	defective (98.4%)	3.9 ms	cpu
2026-09-19 22:14:30	defective (98.4%)	4.3 ms	cpu
2026-09-19 22:14:30	defective (98.4%)	4.3 ms	cpu
2026-09-19 22:14:29	defective (98.4%)	4.8 ms	cpu
2026-09-19 22:14:29	defective (98.4%)	4.9 ms	cpu
2026-09-19 22:14:29	defective (98.4%)	4.3 ms	cpu
2026-09-19 22:14:29	defective (98.4%)	4.3 ms	cpu
2026-09-19 22:14:29	defective (98.4%)	4.2 ms	cpu
2026-09-19 22:14:28	defective (98.4%)	5.6 ms	cpu
2026-09-19 22:14:28	defective (98.4%)	4.2 ms	cpu
2026-09-19 22:14:28	defective (98.4%)	4.8 ms	cpu
2026-09-19 22:14:28	defective (98.4%)	4.6 ms	cpu
2026-09-19 22:14:28	defective (98.4%)	4.4 ms	cpu
2026-09-19 22:14:28	defective (98.4%)	4.4 ms	cpu

Result: **DEFECTIVE (98.4%)**

Latency: 4.3 ms [CPU]

Run Timing Comparison (local CPU)

Run 1: 4.5 ms | Run 2: 3.6 ms (both local CPU — see note above)

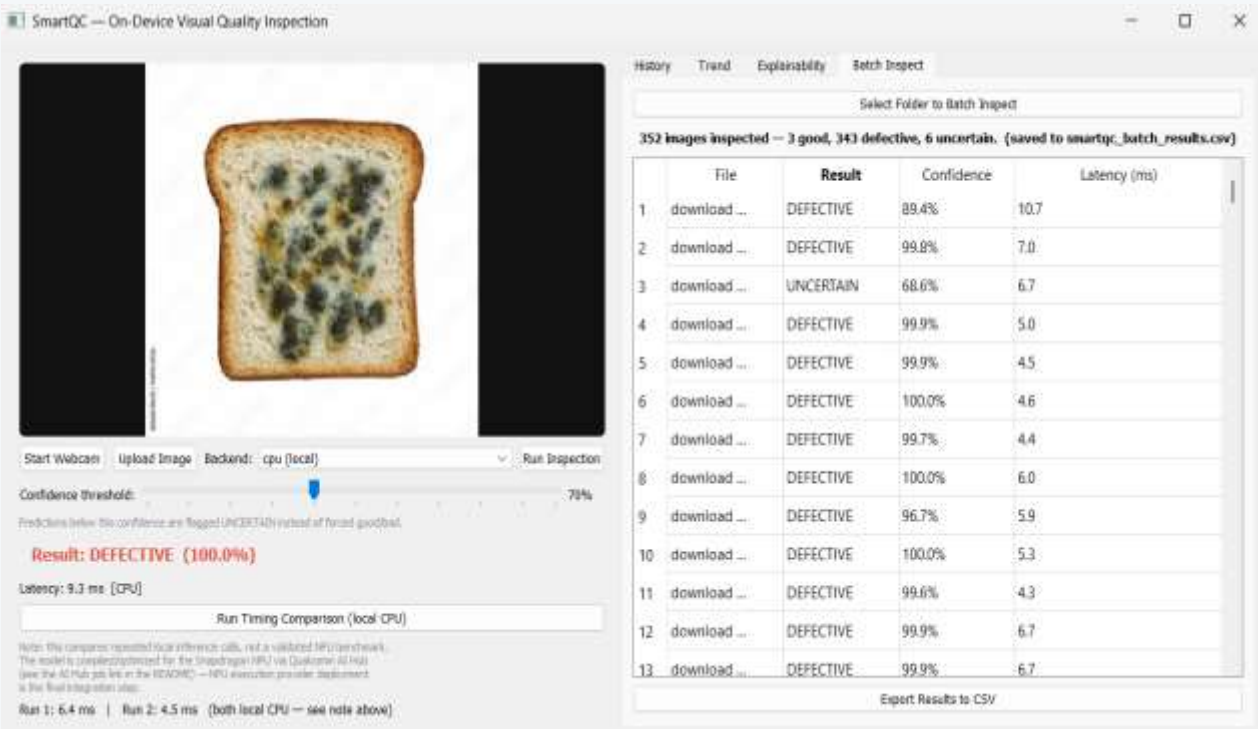
Note: this compares repeated local inference calls, not a validated NPU benchmark. The model is compiled/optimized for the Snapdragon NPU via Qualcomm AI Hub (see the AI Hub job link in the README) — NPU execution provider deployment is the final integration step.

DEFECTIVE — 98.4% confidence

REAL-WORLD VALIDATION

Batch-tested against 352 uncurated images

Beyond the curated 129-image validation set, SmartQC's batch mode was run against a larger, unsorted folder of 352 real-world mold-bread images — a messier, more realistic test than a clean benchmark.



352
Images batch-processed

343
Correctly flagged
DEFECTIVE

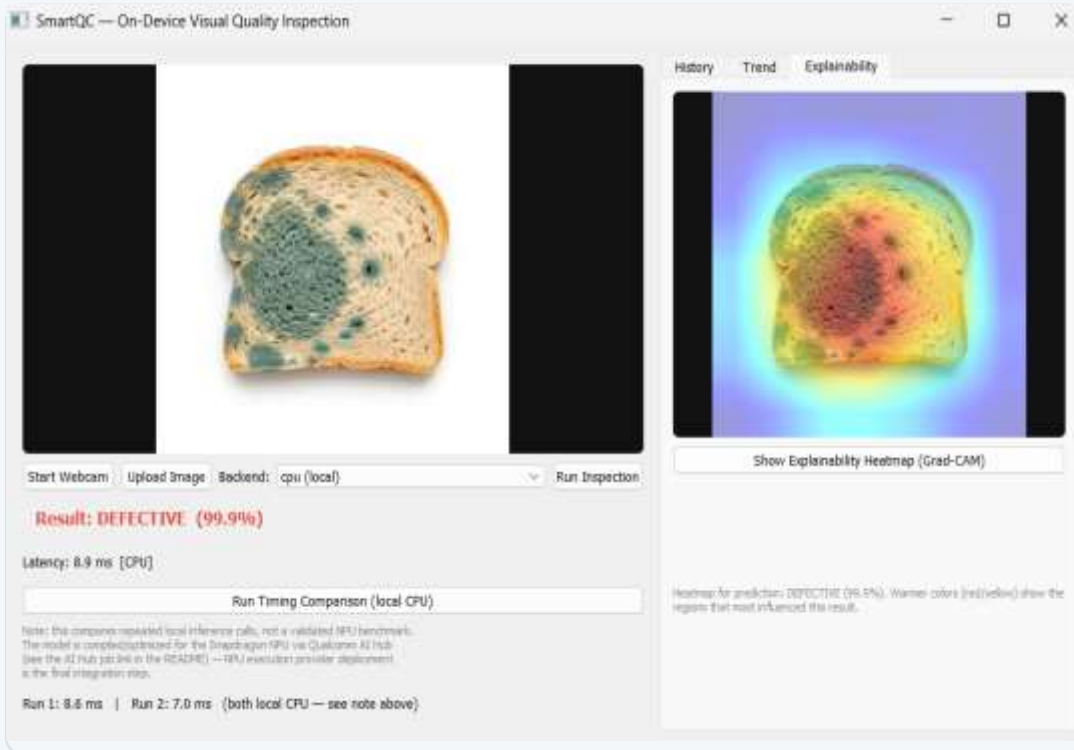
6
Flagged UNCERTAIN
(below threshold)

3
Misclassified (edge cases)

97.4% correct on an uncurated real-world batch — and the confidence-threshold feature caught the ambiguous cases instead of silently guessing wrong.

BEYOND CLASSIFICATION

Explainability: showing WHY, not just what



Grad-CAM heatmap

Highlights exactly which pixels drove the model's decision — warmer colors (red/yellow) show the highest-influence regions.

Builds operator trust

A QC operator can visually verify the model is looking at the actual defect, not an unrelated artifact.

Runs on the trained model directly

Computed via PyTorch alongside the ONNX/AI-Hub inference path — an explainability layer, not a replacement for it.

WHAT'S NEXT

Roadmap

1

Live QNN NPU deployment

Wire up the QNN Execution Provider on physical Snapdragon hardware for measured on-device NPU latency.

2

Multi-product support

Extend beyond bread to additional inspection domains (produce, packaging, components).

3

Multi-class defects

Move beyond good/bad to specific defect types — mold, staleness, physical damage.

4

Fleet / multi-station rollout

Central dashboard aggregating results across multiple inspection stations.

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SmartQC

On-device inspection, privately and instantly, on Snapdragon.

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Thank You